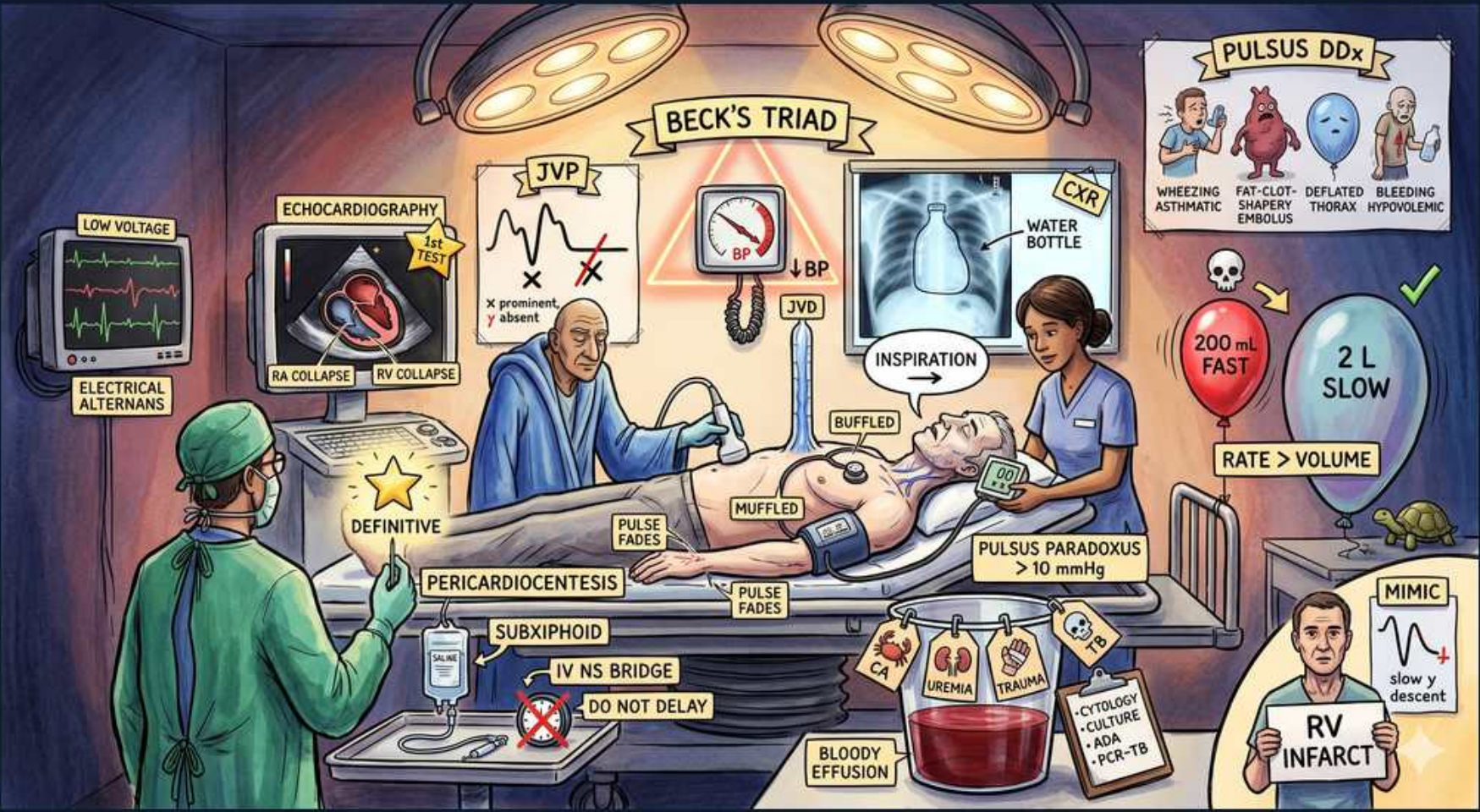


Pericardial Disease — Cardiac Tamponade



1. Beck's triad

WHAT YOU SEE

Banner 'BECK'S TRIAD' over the OR table

WHAT IT ENCODES

Beck's triad = hypotension + muffled heart sounds + elevated JVP (distended neck veins). It reflects the final common pathway of tamponade: pericardial fluid raises intrapericardial pressure above cardiac chamber diastolic pressure, so the heart cannot fill. The full triad is actually present in a MINORITY of patients (especially acute tamponade) — its absence does NOT rule out tamponade. Mnemonic: '3 D's' — Distant heart sounds, Distended neck veins, Decreased arterial pressure.

BOARD TRAP

Classic stem: hypotensive patient days after MI, cardiac surgery, or central-line placement (procedure-related effusion). The wrong move is to attribute hypotension to sepsis or pump failure and give pressors/diuretics — get a STAT echo, because the diagnosis is mechanical and reversible.

2. Hypotension ↓BP

WHAT YOU SEE

Blood-pressure gauge with red arrow, '↓BP'

WHAT IT ENCODES

Hypotension arises because impaired diastolic filling drops stroke volume; the body initially compensates with tachycardia and vasoconstriction, so blood pressure may be borderline before it crashes. As intrapericardial pressure rises, cardiac output becomes critically preload-dependent. Narrow pulse pressure precedes frank hypotension.

BOARD TRAP

Don't be reassured by a 'normal' BP early on — compensated tamponade can maintain pressure until sudden decompensation. Tachycardia + narrowing pulse pressure + rising JVP is the tip-off; treating empirically with beta-blockers (to 'control the tachycardia') removes the only compensatory mechanism and can precipitate arrest.

3. Muffled heart sounds

WHAT YOU SEE

'MUFFLED' label near the patient's chest

WHAT IT ENCODES

Muffled / distant heart sounds occur because the fluid layer insulates and dampens transmission of sound from the heart to the chest wall. This is the least sensitive component of Beck's triad and is hard to appreciate in a noisy ED or a large-volume chronic effusion.

BOARD TRAP

Distant heart sounds are easily confused with COPD/emphysema (hyperinflated lungs) or obesity. Don't anchor on a single auscultatory finding — pair it with JVD and pulsus paradoxus, and confirm with echo rather than relying on the muffled sounds alone.

4. Elevated JVP

WHAT YOU SEE

'JVP' card with prominent x descent, absent y

WHAT IT ENCODES

Elevated JVP with a prominent x descent and an absent/blunted y descent. The y descent is lost because once the pericardium is full, the ventricle cannot accept blood in early diastole (no rapid ventricular filling). Contrast with constrictive pericarditis, which has a prominent y descent ('M' or 'W' waveform). This waveform distinction is a classic board discriminator.

BOARD TRAP

Kussmaul sign (paradoxical RISE in JVP with inspiration) is NOT classic for tamponade — it points to constrictive pericarditis (and also RV infarct, restrictive cardiomyopathy). Picking 'Kussmaul sign' as the tamponade finding is the trap; tamponade shows pulsus paradoxus and a blunted y descent instead.

5. Pulsus paradoxus >10

WHAT YOU SEE

'PULSUS PARADOXUS >10 mmHg' label

WHAT IT ENCODES

Pulsus paradoxus = an exaggerated inspiratory fall in systolic BP >10 mmHg (normal is <10). Mechanism: inspiration increases venous return to the RV, which bulges the interventricular septum leftward into the fixed-volume pericardial space, compromising LV filling and dropping LV stroke volume. Measure by deflating the cuff slowly: the gap between when Korotkoff sounds appear only in expiration and when they appear throughout the cycle.

BOARD TRAP

Pulsus paradoxus is NOT specific to tamponade and can be ABSENT in tamponade when there is a coexisting ASD, severe aortic regurgitation, marked LV dysfunction, or pre-existing elevated LV diastolic pressure. So 'no pulsus paradoxus' does not exclude tamponade — don't use its absence to rule out.

6. Electrical alternans

WHAT YOU SEE

ECG monitor 'ELECTRICAL ALTERNANS', low voltage

WHAT IT ENCODES

Electrical alternans = beat-to-beat alternation in QRS amplitude (and sometimes P/T waves) caused by the heart physically swinging within a large pericardial effusion, changing the QRS vector each beat. Combined with low voltage and sinus tachycardia, this triad on ECG is highly suggestive of a large effusion / tamponade.

BOARD TRAP

Electrical alternans is SPECIFIC but INSENSITIVE — most tamponade has no alternans, so its absence means nothing. Also don't confuse it with the rate-related QRS alternans seen in SVT. It indicates a large effusion but not necessarily hemodynamic tamponade (which is a clinical/echo diagnosis).

7. Low voltage

WHAT YOU SEE

'LOW VOLTAGE' label on the ECG monitor

WHAT IT ENCODES

Low-voltage QRS (limb leads <5 mm, precordial <10 mm) results from the pericardial fluid attenuating/short-circuiting the electrical signal before it reaches surface electrodes. Other causes of low voltage include obesity, COPD, large pleural effusions, infiltrative disease (amyloid), and hypothyroidism (myxedema).

BOARD TRAP

Low voltage is nonspecific — don't diagnose tamponade off the ECG alone. A pericardial effusion is an ANATOMIC finding; tamponade is a HEMODYNAMIC/clinical diagnosis. The board answer for confirming hemodynamic compromise is echocardiography (chamber collapse), not the ECG.

8. Echo diastolic collapse

WHAT YOU SEE

Echo screen 'RA collapse / RV collapse'

WHAT IT ENCODES

Echocardiography is the diagnostic test of choice. Findings: right ATRIAL systolic collapse (early, very sensitive), right VENTRICULAR diastolic collapse (more specific for hemodynamically significant tamponade), a dilated IVC with <50% inspiratory collapse (elevated RA pressure), and exaggerated respiratory variation in transmitral/transtricuspid inflow (Doppler correlate of pulsus paradoxus, >25-40%).

BOARD TRAP

Tamponade is ultimately a CLINICAL diagnosis supported by echo — a large effusion WITHOUT chamber collapse or hemodynamic compromise does NOT require emergent drainage. Conversely, a hemodynamically unstable patient with classic findings should not have drainage delayed to obtain a 'perfect' echo. Don't drain every effusion you see.

9. Water-bottle CXR

WHAT YOU SEE

CXR labeled 'WATER BOTTLE' silhouette

WHAT IT ENCODES

A globular, symmetric 'water-bottle' cardiac silhouette with clear lung fields suggests a large pericardial effusion (requires ~200 mL of fluid to be visible). The clear lungs help distinguish from CHF (which has pulmonary edema/cephalization).

BOARD TRAP

CXR is neither sensitive nor specific and is NEVER required to diagnose or treat tamponade — a normal cardiac silhouette is common in ACUTE tamponade (small rapidly accumulating effusion). Waiting for or relying on the CXR is the wrong answer; echo is the confirmatory test.

10. Pericardiocentesis = tx

WHAT YOU SEE

Star 'DEFINITIVE — PERICARDIOCENTESIS, SUBXIPHOID'

WHAT IT ENCODES

Definitive treatment is urgent removal of pericardial fluid — echo-guided pericardiocentesis (commonly subxiphoid approach) for most cases. Removing even a small volume can dramatically improve hemodynamics because of the steep pericardial pressure-volume curve. Surgical drainage (pericardial window) is preferred for traumatic hemopericardium, purulent effusion, or recurrent malignant effusion.

BOARD TRAP

Do NOT give diuretics or vasodilators (nitrates) — they drop preload and can cause cardiovascular collapse in a preload-dependent heart. Choosing 'IV furosemide' for the hypotensive patient with JVD is the classic trap (they confuse it with CHF). The answer is drainage. Type A aortic dissection with hemopericardium is a special exception — pericardiocentesis can worsen the dissection, so emergent SURGERY is preferred.

11. IV fluids bridge

WHAT YOU SEE

IV bag 'IV NS BRIDGE — DO NOT DELAY'

WHAT IT ENCODES

IV fluids (e.g., a 500 mL normal saline bolus) are a temporizing measure that boosts RV preload to help maintain filling while you arrange definitive drainage. Their benefit is modest and mainly helps the hypovolemic/normovolemic patient — they buy minutes, not a cure.

BOARD TRAP

Fluids are a BRIDGE, never the definitive treatment, and can be counterproductive in a euvolemic/hypervolemic patient (overfilling further raises intrapericardial pressure). Picking 'aggressive IV fluid resuscitation' as the definitive management instead of drainage is the trap — the answer remains pericardiocentesis.

12. Rate > volume

WHAT YOU SEE

'RATE > VOLUME', 200 mL fast vs 2 L slow tortoise

WHAT IT ENCODES

The RATE of accumulation matters more than the absolute volume because of the pericardium's nonlinear compliance. A rapid accumulation of only ~150-200 mL (e.g., trauma, ventricular free-wall rupture post-MI, post-procedure) can cause acute tamponade, whereas a slowly accumulating chronic effusion (e.g., malignancy, hypothyroidism) can reach 1-2 L+ before the stretched pericardium decompensates.

BOARD TRAP

Don't dismiss a 'small' effusion on echo — if it accumulated fast and the patient is unstable, it can be lethal. Conversely a huge chronic effusion in a stable, compensated patient may not be tamponade. Volume alone does not equal tamponade; hemodynamics do.

13. Pulsus DDx

WHAT YOU SEE

'PULSUS DDx': asthma, embolus, tension PTX, hypovolemia

WHAT IT ENCODES

Pulsus paradoxus is not exclusive to tamponade. Other causes: severe asthma/COPD exacerbation (large negative intrathoracic pressure swings), massive pulmonary embolism, tension pneumothorax, hypovolemic/hemorrhagic shock, and constrictive pericarditis (variably). Recognizing pulsus in these contexts changes the management entirely.

BOARD TRAP

A patient with severe asthma and pulsus paradoxus does NOT have tamponade — the trap is reflexively ordering pericardiocentesis. Anchor on the clinical picture: the asthmatic needs bronchodilators, the tension PTX needs needle decompression. Pulsus is a clue, not a diagnosis.

14. RV infarct mimic

WHAT YOU SEE

'RV INFARCT' mimic, slow y descent

WHAT IT ENCODES

RV infarction (usually from a proximal RCA occlusion, accompanying inferior STEMI) mimics tamponade with hypotension + elevated JVP + clear lungs. Distinguish via ECG (inferior STEMI with ST elevation in right-sided lead V4R), echo (RV dilation/hypokinesis without effusion), and the JVP waveform. Treatment is OPPOSITE in spirit: RV infarct needs aggressive IV fluids to maintain preload, and you must AVOID nitrates.

BOARD TRAP

Both RV infarct and tamponade present with hypotension + JVD + clear lungs, but the management diverges — don't reach for pericardiocentesis in an RV infarct (there's no effusion to drain). The discriminator is the echo (effusion + chamber collapse vs RV dysfunction) and the inferior STEMI / V4R ST elevation pointing to RV infarct.
